

Breast and Cervical Cancer Surveillance, United States, 1973-1987

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Summary

Breast and cervical cancer incidence and mortality rates were reviewed for the period 1973-1987. For breast cancer, mortality has been relatively stable, increasing from 26.9/100,000 women in 1973 to 27.1 in 1987. Alternatively, data from the National Cancer Institute's Surveillance, Epidemiology, End Results Program (SEER) showed a 36% increase in the incidence of this malignancy over the same period. In 1987, overall incidence of invasive breast cancer was 111.9/100,000 women. White women experienced lower overall mortality rates and higher overall incidence than black women; however, these differences varied by age. Examination of breast cancer incidence by stage of disease at diagnosis revealed that rates for distant and regional disease have remained relatively stable since 1973. In contrast, rates of localized and in situ cancers exhibited an increase in the 1980s that may correspond to increased use of mammography in this country.

The rate of decline in cervical cancer incidence and mortality has slowed in recent years. In 1987, 3.0 cervical cancer deaths/100,000 women occurred. SEER incidence for invasive disease for that year was 8.2/100,000. Rates varied by race, age, state, and stage of disease. In general, black women experienced much higher incidence and mortality from invasive cervical cancer than white women. For both races, rates of in situ disease were highest among young women and decreased rapidly with age. Rates of in situ cervical cancer were consistently higher than rates of invasive cancer for the time period studied.

INTRODUCTION

Use of early detection techniques has directly affected breast and cervical cancer incidence and mortality rates. The decline in cervical cancer rates from the 1950s through the 1970s has been attributed to widespread implementation of the Papanicolaou (Pap) smear (1). Likewise, the increase in breast cancer incidence over the past decade has been attributed, in part, to greater use of mammography (2,3). In 1990, Congress passed the Breast and Cervical Cancer Mortality Prevention Act (Public Law 101-354). This Act led to the creation of a CDC program for awarding funds to states for establishing comprehensive breast and cervical cancer control programs. These programs emphasize the development and implementation of appropriate screening, follow-up, and treatment programs for these two cancers,

specifically among low-income and minority women. To aid planning and to facilitate future evaluation of these programs nationally, this investigation reviewed trends in both incidence and mortality from these malignancies for the period 1973-1987. Trends in screening prevalence are reviewed in the companion article in this surveillance report (Ackermann SP, Brackbill RM, Bewerse BA, Cheal NE, Sanderson LM: Cancer screening behaviors among U.S. women: breast cancer, 1987-1989, and cervical cancer, 1988-1989. MMWR 1992;41[No. SS-2]:17-34).

METHODS

Mortality

Deaths from breast cancer (*International Classification of Diseases, Adapted, Ninth Revision* [ICD-9] 174.0-174.9) and from cervical cancer (ICD-9 180.0-180.9) were identified from public-use files compiled by the National Center for Health Statistics (NCHS), CDC (4,5). Underlying cause of death was used. Denominators for rate calculations were obtained from intercensal estimates. These population estimates are for the total female population, as well as for white and black women. Intercensal estimates for other specific racial/ethnic groups were not available on the public-use tapes (6).

Mortality rates were directly standardized to the age distribution of the 1970 U.S. population. Analyses by race, state, and calendar year are presented, as well as age-specific mortality rates. Small numbers resulting from stratification of the data across several variables produced some rates that were statistically unstable. This was particularly true for cervical cancer mortality. Therefore, for a few analyses, data from several calendar years were combined to increase precision.

Incidence

Incidence data were obtained from public-use tapes provided by the Surveillance, Epidemiology, End Results Program (SEER) of the National Cancer Institute (NCI) (8). This program has collected information on newly diagnosed cancers since 1973 and covers about 10% of the U.S. population. The public-use file contains data through 1987 from the initial SEER reporting centers (San Francisco-Oakland and standard metropolitan statistical area, Connecticut, metropolitan Detroit, Hawaii, Iowa, New Mexico, and Utah) as well as Seattle (Puget Sound [added in 1974]) and metropolitan Atlanta (added in 1975). Breast cancer cases were defined as tumors with primary sites of codes 740-749 (ICD for Oncology, 1976). The codes 800-809 were used to identify cervical cancers. All lymphomas (morphology codes 9590-9980) were excluded from the case groups.

The annual population estimates for the geographic areas covered by the SEER registries were also provided by NCI. These estimates are based on data from the U.S. Department of the Census and serve as the denominators for incidence calculations. As in the mortality analyses, both age-specific and age-adjusted incidence were calculated for invasive disease. Also examined was the stage of disease at diagnosis, classified as in situ, local, regional, distant, or unknown. This variable is of particular

interest in evaluating breast and cervical cancer detection programs because one would expect changes in the distribution of stage over time if cancer screening levels changed.

RESULTS

Breast Cancer

Breast cancer is the most commonly diagnosed cancer and the second leading cause of cancer death among American women. In 1987, about 41,000 women died from this malignancy. The mortality rate was 27.1 deaths/100,000 women (Figure 1). Mortality has been relatively stable since 1973, increasing only 1%. Alternatively, a 36% increase in the incidence of invasive breast cancer occurred over the same period. Most of this increase occurred in the past decade: between 1980 and 1987, the overall incidence of breast cancer rose from 84.8/100,000 to 111.9/100,000 women. When these rates were stratified by race, white women experienced a greater increase in overall incidence during the 1980s. For white women, rates rose from 86.7/100,000 women in 1980 to 115.9/100,000 women in 1987. Among black women, rates over the same time period rose from 73.3 to 90.9/100,000 women.

Breast cancer mortality rates varied by state. Average annual age-adjusted breast cancer mortality rates for all races, whites, and blacks in 1986-1987 are presented (Table 1). No rate is presented for states whose black or white female population was less than 75,000. Overall, mortality ranged from 18.9/100,000 in Hawaii to 32.6/100,000 in Vermont. In some states, differences between rates for "all races" are a reflection of differences in racial/ethnic composition. Thus, the reported number of deaths among various racial/ethnic groups as well as the distribution of these groups in the population should be reviewed to help interpret rate differences among states.

In general, white women experienced lower overall mortality rates for breast cancer than black women (27.1 versus 30.3/100,000 women in 1987). However, differences between whites and blacks varied by age (Figure 2). Average annual age-specific rates for the years 1983 to 1987 indicated that white women ages ≥ 60 experienced higher mortality than black women in this age range. In contrast to their mortality rate, white women have higher overall incidence than black women. However, black women < 40 years of age experienced slightly higher incidence than white women in corresponding age strata. This relationship has been relatively consistent since 1973. For both races, incidence decreased in the oldest age group.

Trends in breast cancer mortality for 1973-1987 were examined for whites and blacks by calendar year and 10-year age groups (Figure 3). Overall, mortality has been relatively stable among white women (data not shown). However, since 1980, small increases in mortality rates have occurred for white women ≥ 60 years: rates rose 5.3% among those ages 60-69 years, 11.5% among those ages 70-79 years, and 8.3% among those ages ≥ 80 . In the same period, small decreases in mortality ($< 1\%$ to about 8%) were observed for each age group < 60 years. In contrast, breast cancer mortality rates among black women have increased for all age groups since 1980. Moreover, black women ages ≥ 60 exhibited higher increased mortality over time than white women in comparable age groups: between 1980 and 1987, rates for black women rose 17.2% for those ages 60-69 years, 17.9% for those ages 70-79 years, and 17.8% for those ages ≥ 80 years.

SEER incidence data, stratified by race, calendar year, and 10-year age group, are provided (Figure 4). Small numbers for black women necessitate caution in interpret-

ing these rates. As previously noted, breast cancer incidence for whites and blacks has increased over time. For both races, the largest increases are seen among women diagnosed at ages ≥ 60 years.

Rates of in situ and localized cancers showed an increase in the 1980s (Figure 5) that may correspond to increased use of mammography in this country. Except for a small increase in 1974 for regional cancers, rates for distant and regional disease have remained relatively stable. No trend was noted in rates of disease among persons with unknown disease stage (data not shown). Differences were apparent when in situ rates were examined by race, age group, and time period (Table 2). A large increase in the rate of in situ disease was evident among both black and white women for the most recent time period (1983–1987); however, the increase was much more dramatic for white women.

White women ages 40–59 years tended to have a much higher proportion of cancers diagnosed at the in situ stage than women in other age groups. Interestingly, screening tends to be higher at these ages than at others. Given that the prevalence of mammography is increasing and that women in the oldest age groups are less likely to be screened, the results presented in Table 2 are not surprising.

Cervical Cancer

Since 1950, the incidence of invasive cervical cancer in the United States has decreased about 78% and U.S. mortality has decreased about 73% (8). Since the early 1980s, however, the rate of decline in incidence and mortality from this disease has slowed and, in most recent years, may be leveling off (Figure 6). In 1987, 3.0 cervical cancer deaths/100,000 women occurred. The invasive cervical cancer incidence at SEER sites for that year was 8.2/100,000.

As with breast cancer, cervical cancer mortality varied by state. Average annual age-adjusted rates for the years 1986–1987 combined are presented (Table 3). Small numbers preclude calculating rates for black women in several states. Mortality rates for all races combined ranged from 1.6/100,000 in Utah to 6.8/100,000 in the District of Columbia.

Black women experienced much higher incidence and mortality from invasive cervical cancer than white women. In 1987, the death rate was 2.6 times higher among black women (6.8/100,000 versus 2.6/100,000), and disease incidence was twice as high among blacks (15.1/100,000 versus 7.3/100,000). A number of reasons are thought to be involved, such as differences in the prevalence of risk factors for the disease; differences in screening, follow-up, and treatment; and differences in stage of disease at diagnosis (9,10).

Age-specific analyses, which combined data from the years 1983–1987, also revealed significant racial differences in the patterns of disease incidence and mortality (Figure 7). For both races, similar rates of invasive disease were diagnosed among women in age groups 30–34 years or younger. However, although the increase in incidence slowed and leveled for white women after this age, rates among black women continued to increase rapidly through age 44, decreased slightly between the ages of 45 and 54, and then increased again. Mortality among blacks followed a somewhat similar trend. In general, death rates among white women increased with age.

Examination of trends in mortality revealed that, in general, cervical cancer incidence and mortality have decreased over time for all age groups (Figures 8 and 9). Women <35 years have consistently experienced low rates of this malignancy over the past 15 years.

Data from NCI and the Connecticut Tumor Registry indicate that the incidence of carcinoma in situ of the uterine cervix increased between the 1940s and the 1970s (11). Data for all races demonstrate a decrease in the rate of in situ disease after the mid-1970s and a slight increase in more recent years (Figure 10). This recent increase is most likely due to changes in data collection practices at SEER reporting centers in 1983; preinvasive cervical lesions included in SEER data after this time consist of all cervical intraepithelial neoplasia III (CIN III) cases, a classification that combines carcinomas in situ and severe dysplasias. For both whites and blacks, rates of in situ disease were highest among young women and decreased rapidly with age (Table 4). For each age group <50 years of age, over 60% of all cervical cancers diagnosed were staged as in situ disease. This number may underestimate the actual proportion of in situ diagnoses, because case ascertainment is probably incomplete for this stage of disease. Among women ≥ 50 years, there appear to be greater increases in the proportion of in situ diagnoses for white women than for black women since 1973.

DISCUSSION

The rates presented in this report may be biased by a number of factors. Changes in incidence can reflect differences in case finding and case definition over time (particularly for in situ cervical cancer). For cervical cancers, rates may also be influenced by changes in the proportion of women in the population who have had a hysterectomy, although this proportion is reported to have stabilized since the 1970s (1). Mortality rates are affected by how accurately breast or cervical cancer is reported on the death certificate; for example, rates are influenced by the specificity by which uterine cancer site (corpus; cervix; uterus, not otherwise specified) is recorded. No attempt was made in these analyses to adjust for any of the factors that could have influenced incidence or mortality rates.

Breast cancer and cervical cancer exhibit different patterns of disease. Common to both, however, is the potential that early detection and treatment can offer for successfully altering the natural history of these diseases and thus reducing mortality. The benefits of mammography have been demonstrated in a number of clinical trials (12). In the United States, data from the Health Insurance Plan of New York indicate that after 10 years, use of mammography combined with the clinical breast examination resulted in a 30% reduction in mortality from breast cancer in women screened (12). Currently, the value of breast cancer screening for women ages ≥ 50 years is widely accepted. Although the efficacy of cervical cancer screening using the Pap smear has not been evaluated in clinical trials, the marked decline in cervical cancer incidence and mortality rates testifies to its impact in the United States. Improved survival results from early diagnosis and treatment of breast and cervical cancers. For cases diagnosed between 1981 and 1987, relative survival estimates from SEER (i.e., the observed survival rate adjusted for expected mortality) indicate that at least 77% of patients with invasive breast cancer and 66% of patients with invasive cervical cancer will survive 5 years (8). Survival increases dramatically when malignancies are diagnosed at earlier stages. For example, 92% of breast cancer patients with localized disease survive 5 years. For cervical cancer patients

with localized disease, the 5-year survival is 88%; among women diagnosed with carcinoma in situ of the uterine cervix or with dysplastic lesions, survival is essentially 100% (8).

Current racial differences in mortality may be explained by differences in stage of disease at diagnosis and by differences in survival at given stages. Although these analyses found that both races had a similar proportion of cervical cancers staged as in situ, a higher proportion of invasive disease has been diagnosed at regional and distant stages among black women than among white women. This is also true for invasive breast cancer. Moreover, regardless of stage of diagnosis, black women with either cancer experience poorer survival (8).

Historically, the proportion of black women who have received screening examinations has been lower than that of white women (9). However, the use of both breast and cervical cancer screening has increased among black women in recent years and, for cervical cancer screening, may have surpassed whites (9). Selected results from the Behavioral Risk Factor Surveillance System (BRFSS) are discussed in a separate article in this publication and appear to indicate that comparable proportions of white and black women reported having ever been screened for breast or cervical cancer in the 1989 survey (Ackermann SP, Brackbill RM, Bewerse BA, Cheal NE, Sanderson LM: Cancer screening behaviors among U.S. women: breast cancer, 1987-1989, and cervical cancer, 1988-1989. MMWR 1992;41[No. SS-2]:17-34). However, the results of the 1989 BRFSS, the 1990 Jacob's Institute Survey of mammography utilization, and the 1987 National Health Institute Survey also indicate that screening prevalence is much lower in three segments of the population: low-income women, women with low educational attainment, and older women (for Pap smears) (9,13).

Despite the increasing frequency of cervical and breast cancer screening (especially for black women) (9) and improvements in the management of these neoplasms, 45,319 women died from these malignancies in 1987. The Public Health Service has targeted reducing breast cancer mortality rates almost 8% and cervical cancer mortality rates almost 50% by the year 2000 (13). To achieve these goals, special screening objectives have targeted all women and selected subgroups for whom regular screening has been lacking. In addition to screening, follow-up and treatment of detected abnormalities are essential. All states currently funded by CDC to establish breast and cervical cancer control programs are required to track women with abnormal screening results and to ensure that appropriate follow-up and medical treatment are provided.

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TABLE 1. Breast cancer deaths* and age-adjusted mortality rates†, by state and race — United States, 1986–1987

State	No. deaths		Rate/100,000 women	
	All races	White	Black	White
Alabama	1,195	873	321	23.3
Alaska	73	64	1	31.7
Arizona	1,001	963	24	25.2
Arkansas	707	630	76	23.0
California	8,307	7,502	576	27.8
Colorado	807	778	23	25.1
Connecticut	1,300	1,246	54	30.2
Delaware	249	200	47	29.3
Dist. of Columbia	269	67	201	24.3
Florida	4,576	4,170	401	23.3
Georgia	1,749	1,305	439	25.6
Hawaii	206	82	1	27.7
Idaho	238	234	1	21.8
Illinois	4,192	3,635	545	29.1
Indiana	1,890	1,755	132	27.6
Iowa	1,019	1,008	11	25.8
Kansas	789	747	38	23.7
Kentucky	1,103	1,023	80	24.4
Louisiana	1,131	783	347	22.6
Maine	401	400	1	25.7
Maryland	1,575	1,259	307	29.9
Massachusetts	2,551	2,476	66	30.6
Michigan	3,111	2,734	369	28.7
Minnesota	1,411	1,393	11	27.3
Mississippi	714	486	224	23.7
Missouri	1,832	1,673	158	26.4
Montana	252	248	0	26.9
Nebaska	536	537	8	27.0
Nevada	291	274	11	27.2

*From National Center for Health Statistics mortality tapes, underlying cause of death.

†Adjusted to the 1970 U.S. population.

—Less than 75,000 in denominator.

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TABLE 1. Breast cancer deaths* and age-adjusted mortality rates†, by state and race — United States, 1986–1987 — Continued

State	No. deaths			Rate/100,000 women		
	All races	White	Black	All races	White	Black
New Hampshire	356	354	1	28.7	28.7	—
New Jersey	3,214	2,918	284	31.1	31.4	29.1
New Mexico	378	362	5	24.9	25.9	—
New York	7,513	6,651	815	31.1	31.7	28.7
North Carolina	2,055	1,585	459	26.8	25.6	32.5
North Dakota	200	197	0	24.8	24.8	—
Ohio	4,130	3,759	367	30.0	30.0	31.3
Oklahoma	983	886	71	25.3	25.4	35.7
Oregon	904	883	14	26.4	26.6	—
Pennsylvania	4,974	4,583	387	28.9	28.9	30.6
Rhode Island	424	413	9	30.6	30.7	—
South Carolina	934	695	239	24.6	24.6	25.2
South Dakota	252	244	0	27.6	27.8	—
Tennessee	1,522	1,256	266	25.5	24.5	33.0
Texas	3,916	3,407	490	23.1	22.6	28.5
Utah	319	316	1	22.7	23.0	—
Vermont	210	210	0	32.6	32.8	—
Virginia	1,815	1,486	324	28.0	28.0	29.8
Washington	1,343	1,290	31	25.3	25.7	37.8
West Virginia	678	654	22	27.6	27.6	—
Wisconsin	1,705	1,661	36	27.5	27.5	25.7
Wyoming	130	130	0	30.0	30.6	—
Total	81,430	72,475	8,294	27.2	27.3	29.8

*From National Center for Health Statistics mortality tapes, underlying cause of death.

†Adjusted to the 1970 U.S. population.

—Less than 75,000 in denominator.

TABLE 2. Age-specific incidence and proportion of in situ breast cancers, by age group, race, and time period — SEER*, 1973–1987

In situ rates/100,000 women
Proportion of all breast cancers (%)

Race/age	1973–1977	1978–1982	1983–1987	1973–1977	1978–1982	1983–1987
Whites						
20–29	0.2	0.2	0.3	5.0	5.0	7.0
30–39	2.5	2.1	3.9	6.0	5.0	8.0
40–49	11.9	10.7	24.0	8.0	7.0	14.0
50–59	11.3	10.8	30.5	5.0	5.0	11.0
60–69	10.6	10.5	30.4	3.0	3.0	8.0
70–79	9.6	11.9	28.9	3.0	3.0	6.0
≥80	0.5	9.4	14.5	2.0	3.0	3.0
Blacks						
20–29	0.0	0.2	0.2	2.0	2.0	2.0
30–39	2.4	1.8	3.7	4.0	4.0	6.0
40–49	5.8	10.3	15.1	8.0	8.0	9.0
0–59	6.6	6.2	14.3	3.0	3.0	7.0
60–69	7.3	7.7	21.1	3.0	3.0	7.0
70–79	12.4	15.4	32.9	5.0	5.0	9.0
≥80	—	—	16.8	3.0	3.0	4.0

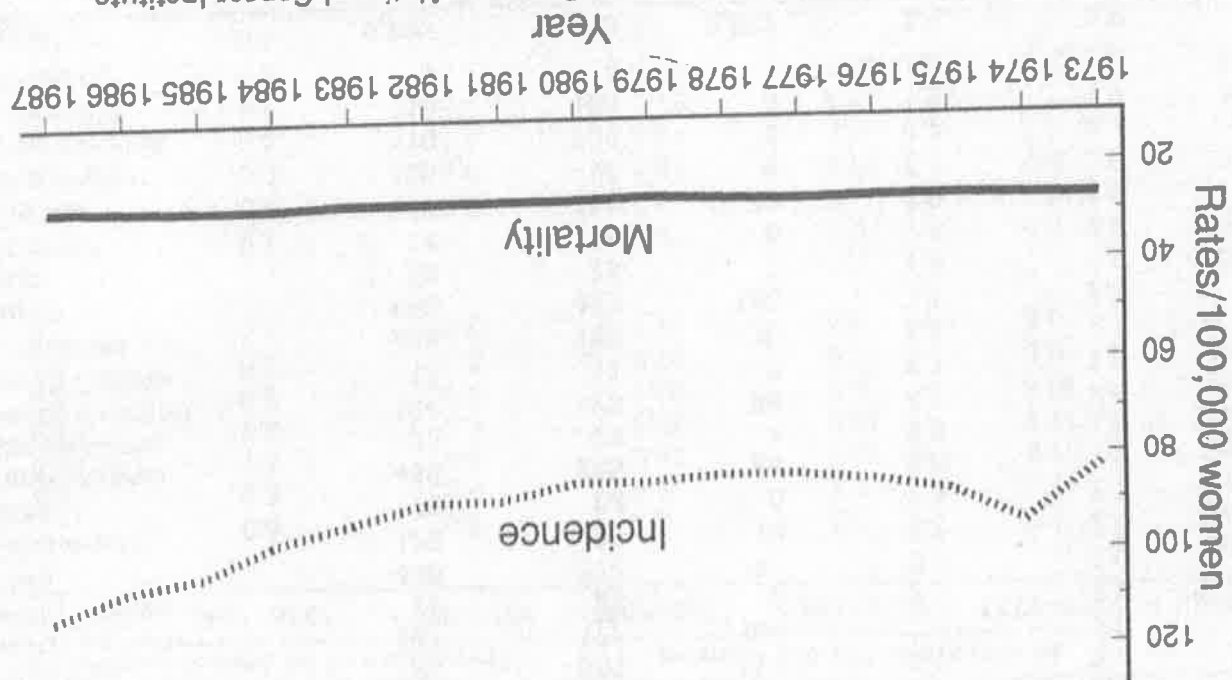
*SEER = Surveillance, Epidemiology, End Results Program, National Cancer Institute.

TABLE 4. Age-specific incidence and proportion of in situ cervical cancers, by age group, race, and time period – SEER*, 1973–1987

Race/age	In situ rates/100,000 women				Proportion of all cervical cancers (%)			
	1973–1977	1978–1982	1983–1987		1973–1977	1978–1982	1983–1987	
Whites								
20–29	81.4	79.2	86.3	93.6	94.3	94.8	94.8	
30–39	105.7	90.3	84.1	86.2	87.3	86.5	86.5	
40–49	47.4	39.8	43.3	70.7	71.8	73.8	73.8	
50–59	20.5	15.2	16.3	47.2	48.0	52.0	52.0	
60–69	15.2	11.7	12.6	36.6	35.6	44.4	44.4	
70–79	9.6	7.8	8.7	28.0	28.3	33.6	33.6	
≥80	3.7	3.9	4.7	11.3	15.9	22.3	22.3	
Blacks								
20–29	169.5	111.3	75.6	94.4	94.2	94.7	94.7	
30–39	174.4	116.6	78.5	85.5	85.2	82.7	82.7	
40–49	82.2	64.4	55.5	65.9	65.0	64.3	64.3	
50–59	41.6	33.8	24.4	42.6	44.4	46.2	46.2	
60–69	33.9	22.9	22.6	36.6	36.0	36.4	36.4	
70–79	34.3	28.2	24.1	30.3	29.0	32.2	32.2	
≥80	21.1	21.5	16.8	19.0	18.6	21.1	21.1	

*SEER = Surveillance, Epidemiology, End Results Program, National Cancer Institute.

FIGURE 1. Breast cancer – SEER* incidence and U.S. mortality rates, 1973–1987



*SEER = Surveillance, Epidemiology, End Results Program, National Cancer Institute.

FIGURE 2. Average annual age-specific SEER* incidence and U.S. mortality rates, by race, 1983-1987

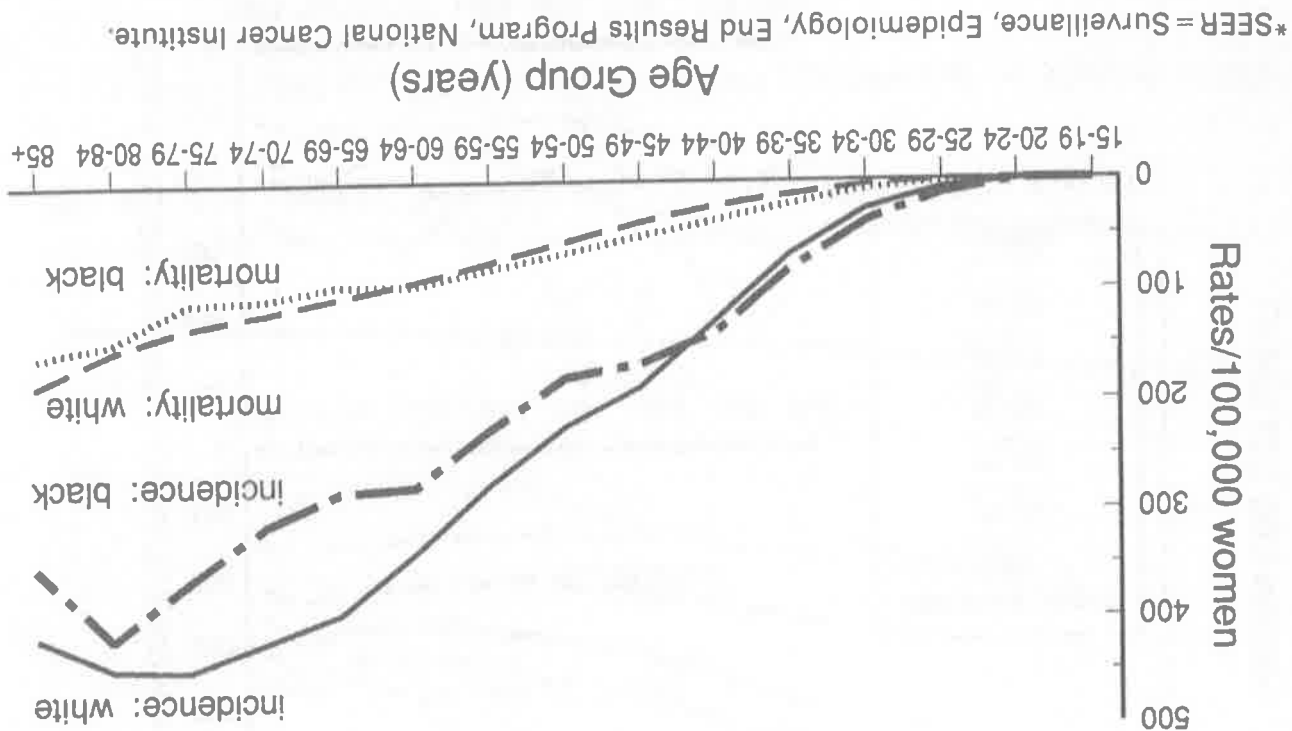


FIGURE 3. Breast cancer mortality rates, by age group and race - United States, 1973-1987

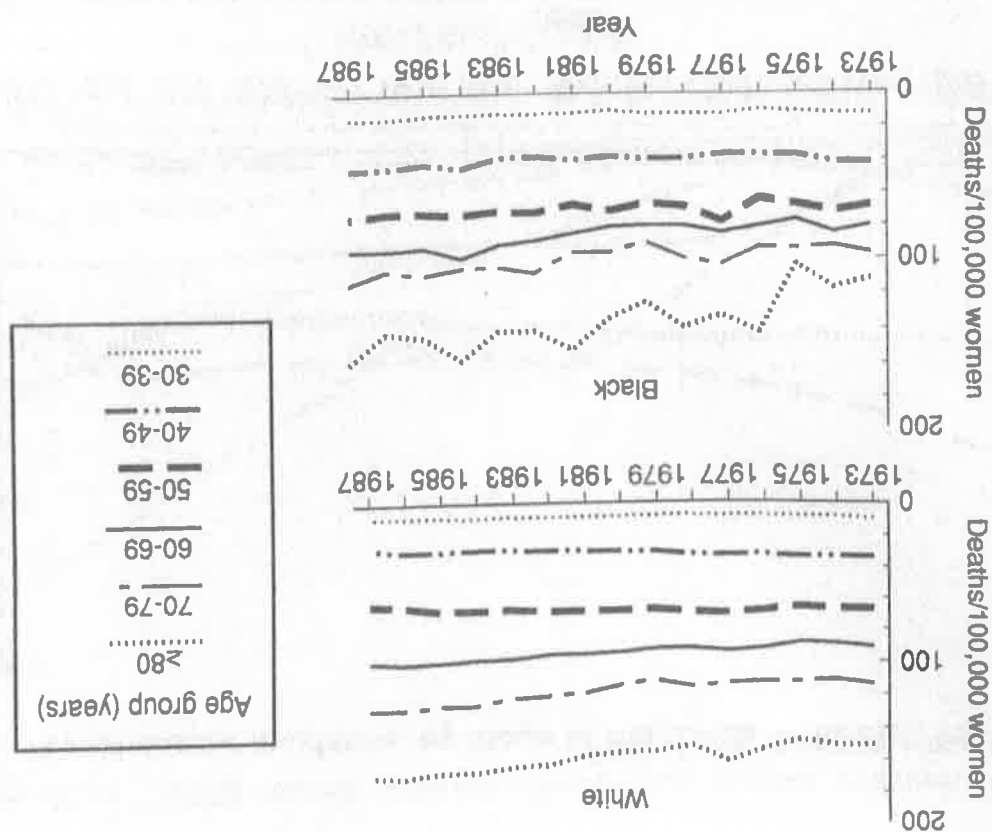
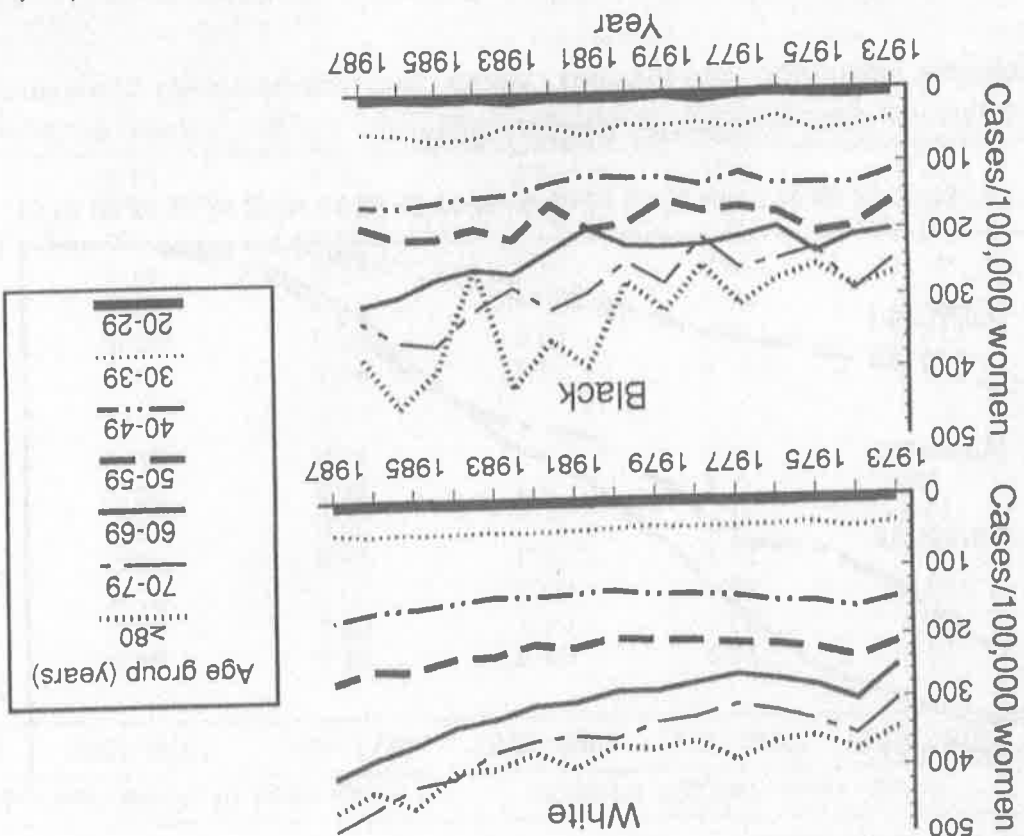
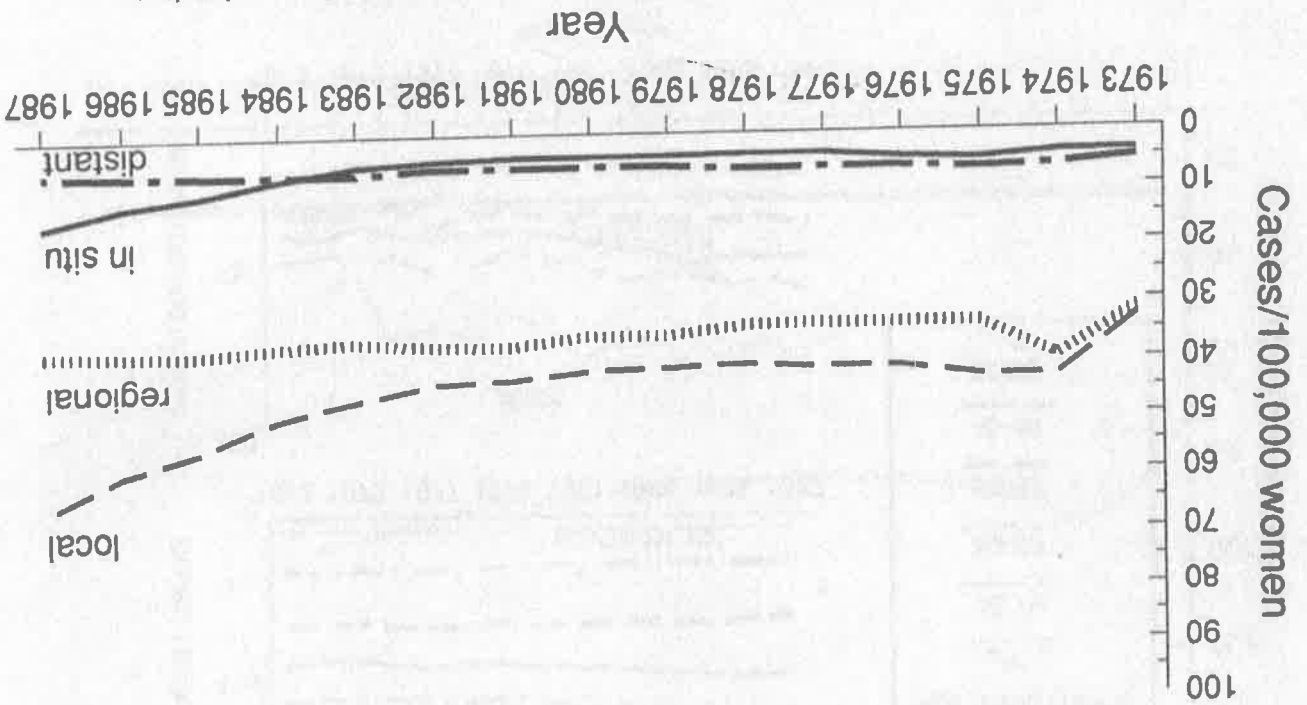


FIGURE 4. Breast cancer incidence, by age group and race - SEER*, 1973-1987



*SEER = Surveillance, Epidemiology, End Results Program, National Cancer Institute.

FIGURE 5. Breast cancer incidence, by stage at diagnosis - SEER*, 1973-1987



*SEER = Surveillance, Epidemiology, End Results Program, National Cancer Institute.

FIGURE 6. Cervical cancer — SEER* incidence and U.S. mortality rates, 1973–1987

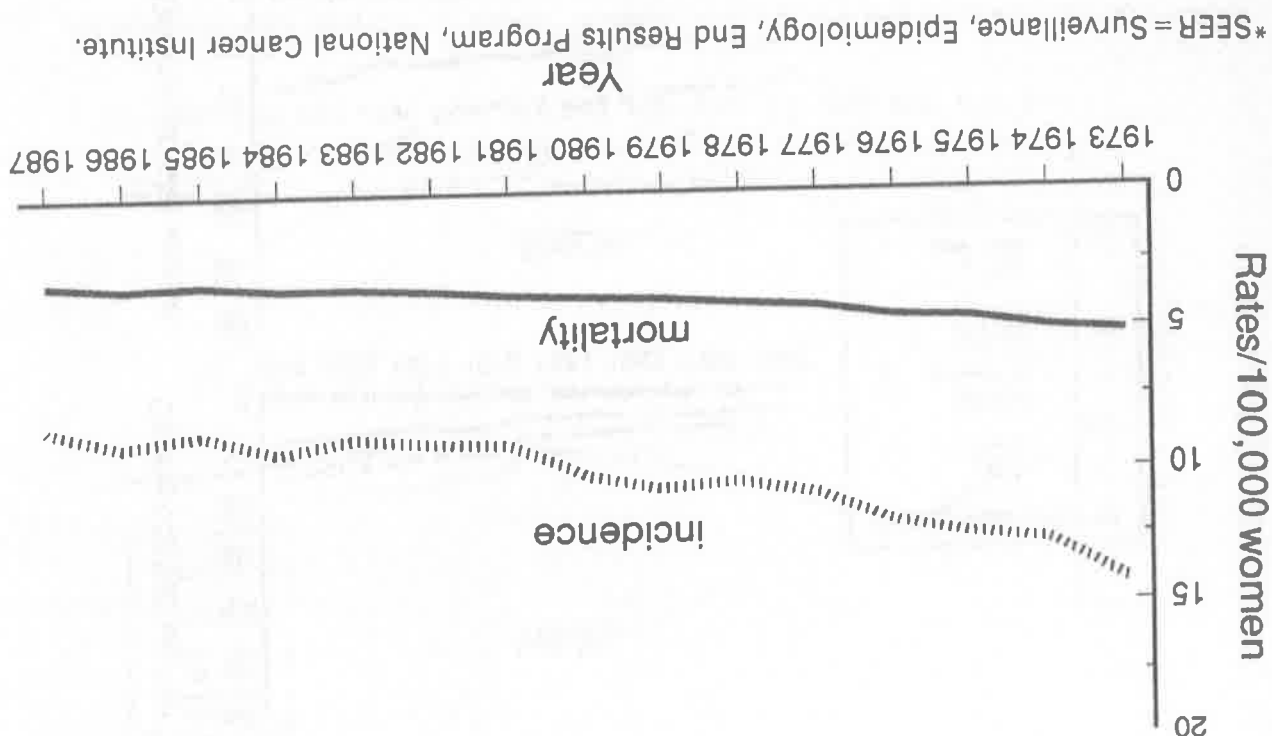


FIGURE 7. Average annual age-specific cervical cancer SEER* incidence and U.S. mortality rates, by race, 1983–1987

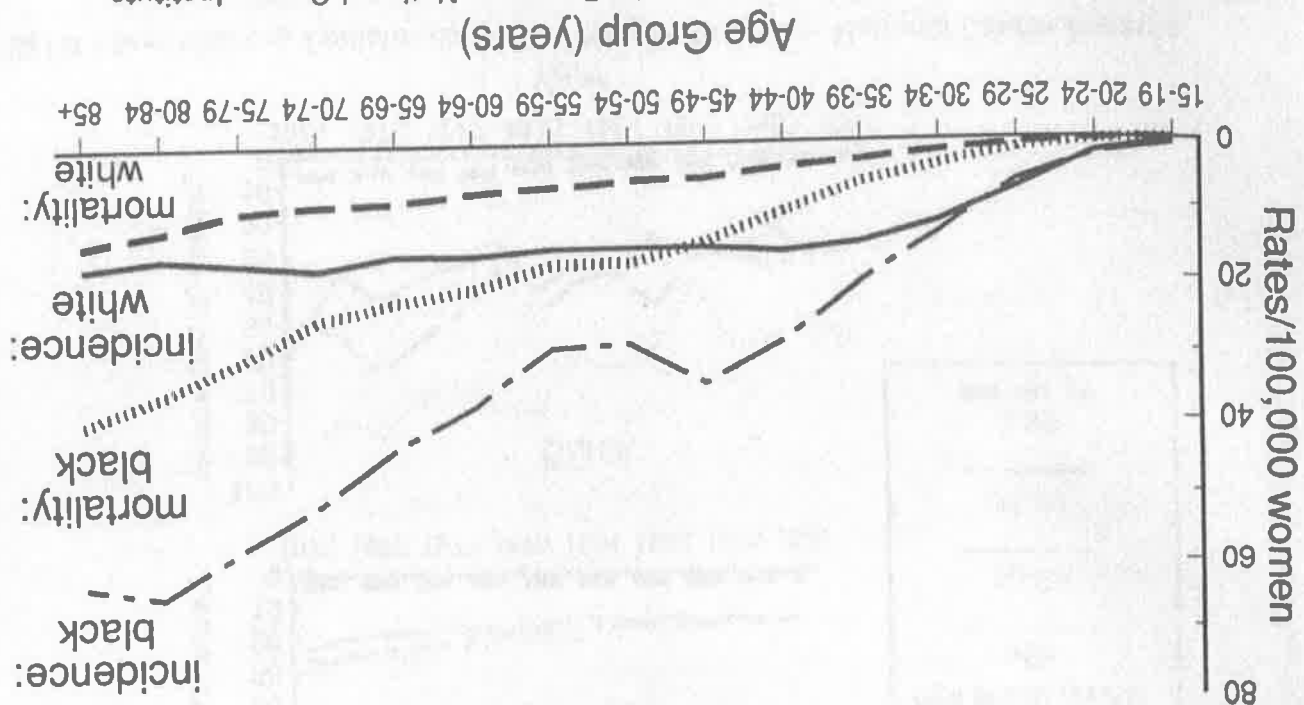


FIGURE 8. Cervical cancer mortality rate, by age group and race — United States, 1973-1987

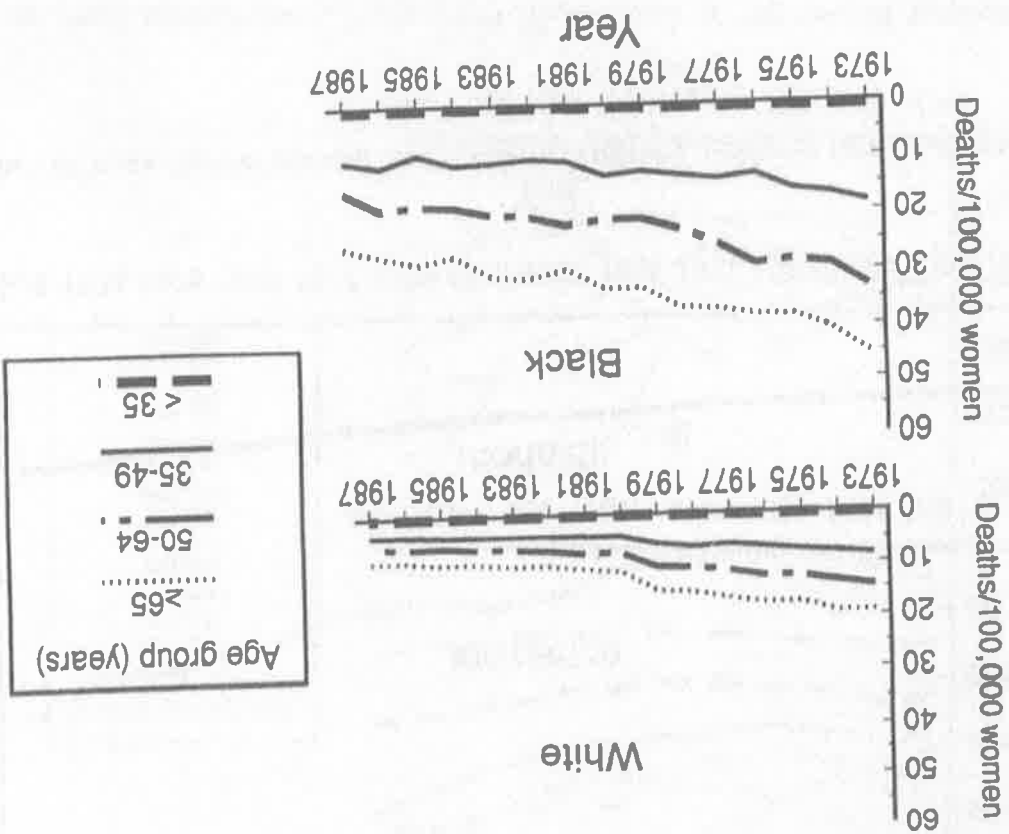
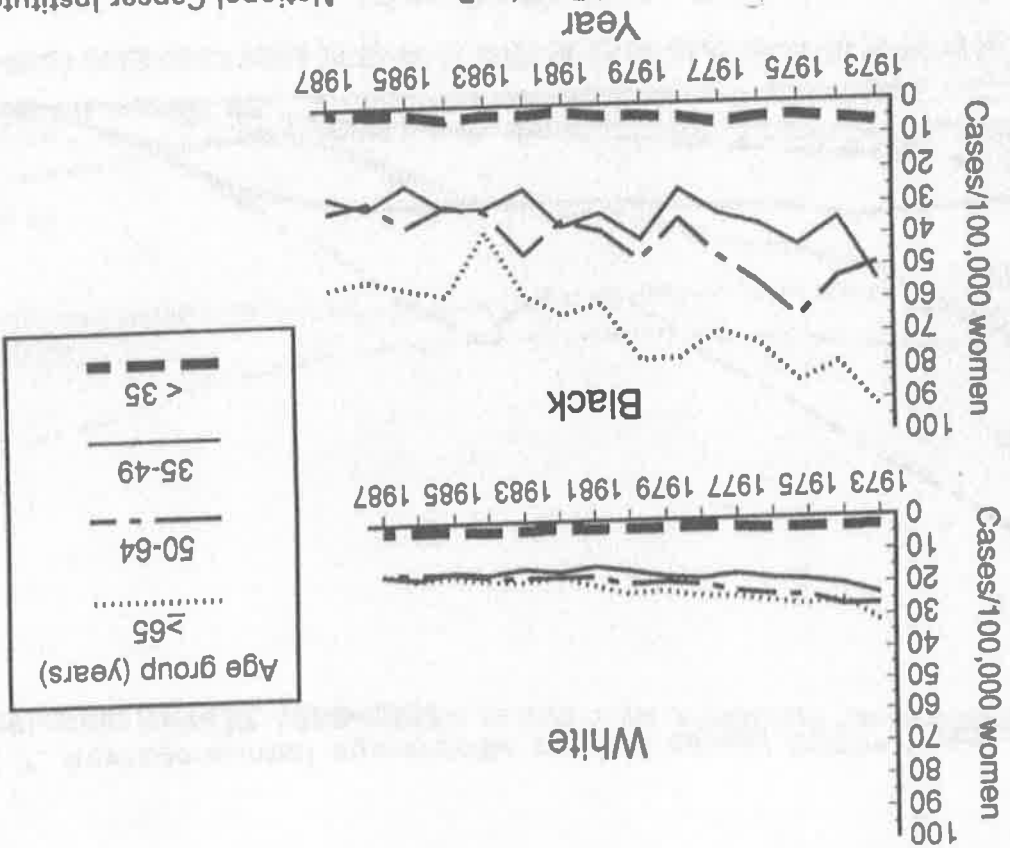


FIGURE 9. Cervical cancer incidence, by age group and race — SEER*, 1973-1987



*SEER = Surveillance, Epidemiology, End Results Program, National Cancer Institute.

FIGURE 10. Cervical cancer incidence, by stage at diagnosis — SEER*, 1973–1987

